The nylon or rayon layers which connect the beads on both sides of the tyre are arranged in the same direction. This type of tyre is called as radial ply tyre. In this type of tyres the stability and comfort are more. However, while negotiating a turn on the bends of the road the stability of the tyre is less. Figure 5.11.1 shows the view of a radial ply tyre.

5.11.2 Cross ply tyre

Nylon or rayon layers that connect the beads on both sides of the tyre are arranged in the opposite (at the angle of 40° to 45°) direction. This type of tyre is called as cross ply type tyre. As the tyre is made the plies with cross winding the tyre gets more strength. A view of the cross ply tyre can be seen in Figure 5.11.2.

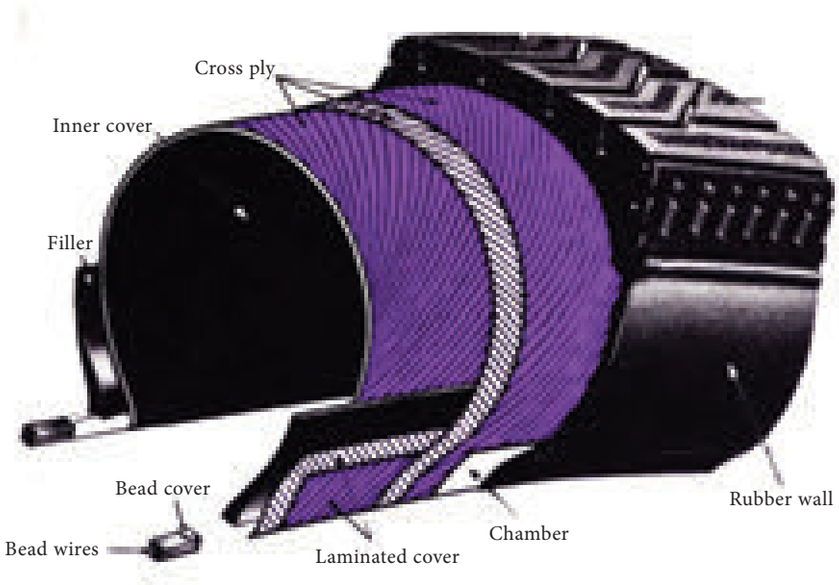


Figure 5.11.2 View of the cross ply tyre

5.11.3 Bias ply tyre

The combination of the tyre construction with radial ply tyre and cross ply tyre is said to be the bias ply tyre. This type of tyre is very strong and has more stability. Figure 5.11.3 shows the concept of bias ply tyre.

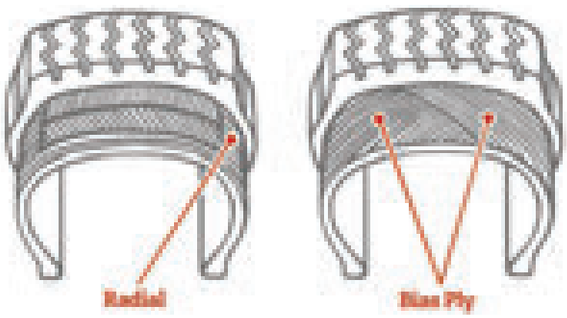


Figure 5.11.3 Bias Ply concept

5.12 Load rating

The tyre's strength, tyre dimensions and it's load carrying capacity are represented by the term called as the load rating.

5.13 Tyre pressure

The air stored at high pressure inside the tyre withstands all the loads

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